

2.2 Design and performance Evaluation of a Chebyshev Type I IIR High Pass Filter using MATLAB

Programming C ode:

```
clc;
clear;
close all;

%% Filter Specifications
Fs = 1000;    % Sampling Frequency (Hz)
Fp = 150;     % Passband Frequency (Hz)
Fst = 100;    % Stopband Frequency (Hz)
Rp = 1;       % Passband Ripple (dB)
Rs = 40;      % Stopband Attenuation (dB)

%% Normalize Frequencies
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

%% Calculate Minimum Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

%% Design Chebyshev Type-I High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');

%% Display Filter Information
disp('Chebyshev Type-I IIR High-Pass Filter');
disp(['Filter Order = ',num2str(n)]);
disp('Numerator Coefficients (b):');
disp(b);
disp('Denominator Coefficients (a):');
disp(a);
```

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%% Frequency Response
figure;
freqz(b,a,1024,Fs);
title('Frequency Response of Chebyshev Type-I High-Pass Filter');

%% Magnitude and Phase Response
[H,f] = freqz(b,a,1024,Fs);
figure;
subplot(2,1,1);
plot(f,20*log10(abs(H)),'b','LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response');
subplot(2,1,2);
plot(f,unwrap(angle(H))*180/pi,'r','LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
title('Phase Response');

%% Impulse Response
figure;
impz(b,a);
grid on;
title('Impulse Response');

%% Pole-Zero Plot
figure;

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zplane(b,a);  
grid on;  
title('Pole-Zero Plot');  
%% Generate Test Signal  
t = 0:1/Fs:1;  
% Low-frequency signal (to be removed)  
x1 = sin(2*pi*50*t);  
% High-frequency signal (to pass)  
x2 = sin(2*pi*250*t);  
% Combined Signal  
x = x1 + x2;  
%% Apply High-Pass Filter  
y = filter(b,a,x);  
%% Input and Output Signals  
figure;  
subplot(2,1,1);  
plot(t,x,'b');  
grid on;  
xlabel('Time (s)');  
ylabel('Amplitude');  
title('Input Signal');  
subplot(2,1,2);  
plot(t,y,'r');  
grid on;  
xlabel('Time (s)');  
ylabel('Amplitude');  
title('Filtered Output Signal');
```

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%% FFT Analysis
N = length(x);
X = abs(fft(x));
Y = abs(fft(y));
faxis = (0:N-1)*(Fs/N);
figure;
subplot(2,1,1);
plot(faxis,X,'b');
xlim([0 500]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Spectrum of Input Signal');
subplot(2,1,2);
plot(faxis,Y,'r');
xlim([0 500]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Spectrum of Filtered Signal');

%% Display Filter Coefficients
fprintf('\nNumerator Coefficients (b):\n');
disp(b);
fprintf('\nDenominator Coefficients (a):\n');
disp(a);

```

OUTPUTS:



